

STA302 Final Project Report

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Section 1: Introduction. (345 words)

Students' first-year academic performance is an outcome connected to later persistence, academic standing, and degree completion. Characteristics observed at university entry may help explain why some students perform better than others during the first year. In this report, we study how first-year academic performance is associated with prior academic achievement, demographic characteristics, and financial support at enrollment.

Our research question is: How is first-year academic performance associated with prior academic achievement, demographic characteristics, and financial support at enrollment? As in Part 1, we focus on five enrollment-time variables motivated by prior literature: admission grade, previous qualification grade, age at enrollment, gender, and scholarship-holder status. We expected higher prior achievement and scholarship support to be associated with better first-year performance, and older age at enrollment to be associated with lower performance.

The data come from the Predict Students' Dropout and Academic Success dataset (Realinho et al., 2021), a student-level administrative dataset from a Portuguese higher education institution. Our goal is interpretability rather than pure prediction. Specifically, we aim to estimate and interpret adjusted associations between first-year grades and characteristics observed before or at the start of university. Because the response contains a substantial point mass at zero, we also anticipated that careful model checking might narrow the final inferential target relative to the original full-sample question.

The literature gives a clear motivation for this question. Wolniak and Engberg (2010) found that prior academic achievement and educational background are important predictors of first-year college performance, which supports including admission grade and previous qualification grade. Newman-Ford, Lloyd, and Thomas (2009) reported that prior academic achievement is a strong predictor of first-year attainment, while age and gender also show meaningful associations. Stater (2009) found that financial aid is positively associated with college GPA, motivating the inclusion of scholarship-holder status.

A multiple linear regression framework is a reasonable starting point because it allows us to estimate adjusted mean differences and adjusted linear trends while holding other predictors fixed. However, diagnostics and model revision remain essential before drawing final conclusions from any regression analysis. This motivation justified our initial full-model specification.

Section 2: Data description. (305 words)

The data come from the Predict Students' Dropout and Academic Success dataset on the UCI Machine Learning Repository (Realinho et al., 2021). The Polytechnic Institute of Portalegre assembled the data from administrative records, national admission records, and macroeconomic indicators. The dataset includes 4,424 students across 17 programs from 2008 to 2019. Although it was originally created to study dropout and academic success, we use it here to explain variation in first-year average grade using enrollment characteristics observed at enrollment.

Our response variable is `grade_year1`, the average of the first-semester and second-semester grade variables in the cleaned dataset. This variable is continuous and has a clear academic interpretation. Most students are concentrated in the mid-to-upper grade range, but there is also a substantial point mass at zero. These zero values are substantively meaningful because they correspond to students who withdrew early or completed no graded coursework. In the full sample, 676 of 4424 students, or 15.3%, have `grade_year1` = 0.

The five predictors carried forward from Part 1 are `admission_grade`, `previous_qualification_grade`, `age_at_enrollment`, `gender`, and `scholarship_holder`. The first three are numerical variables, and the last two are treated as categorical predictors by recoding them as factors. The descriptive summaries show that `age_at_enrollment` is strongly right-skewed, while `grade_year1` is left-skewed because of the large concentration at zero. The histogram, numerical summary table, and frequency table provide the main explanatory data analysis used to motivate the later model-selection work. The descriptive summaries also show that `admission_grade` and `previous_qualification_grade` are centered in the 120 to 130 range with moderate spread, whereas `age_at_enrollment` has a much longer right tail extending to older students. Because `gender` and `scholarship_holder` are binary coded variables, their frequency summaries clarify the reference and comparison categories that appear later in the regression output, coefficient interpretations, and discussion of adjusted group differences throughout the written report sections.

Table 1. Numerical summary of the response and numerical predictors.

Variable	Mean	SD	Min	Max	Missing	Skewness
admission_grade	126.978	14.482	95	190.000	0	0.530
age_at_enrollment	23.265	7.588	17	70.000	0	2.054
grade_year1	10.436	4.819	0	18.284	0	-1.430
previous_qualification_grade	132.613	13.188	95	190.000	0	0.313

Table 2. Frequency summary of the categorical predictors.

Variable	Level	n	Proportion
gender	0	2868	0.648
gender	1	1556	0.352
scholarship_holder	0	3325	0.752
scholarship_holder	1	1099	0.248

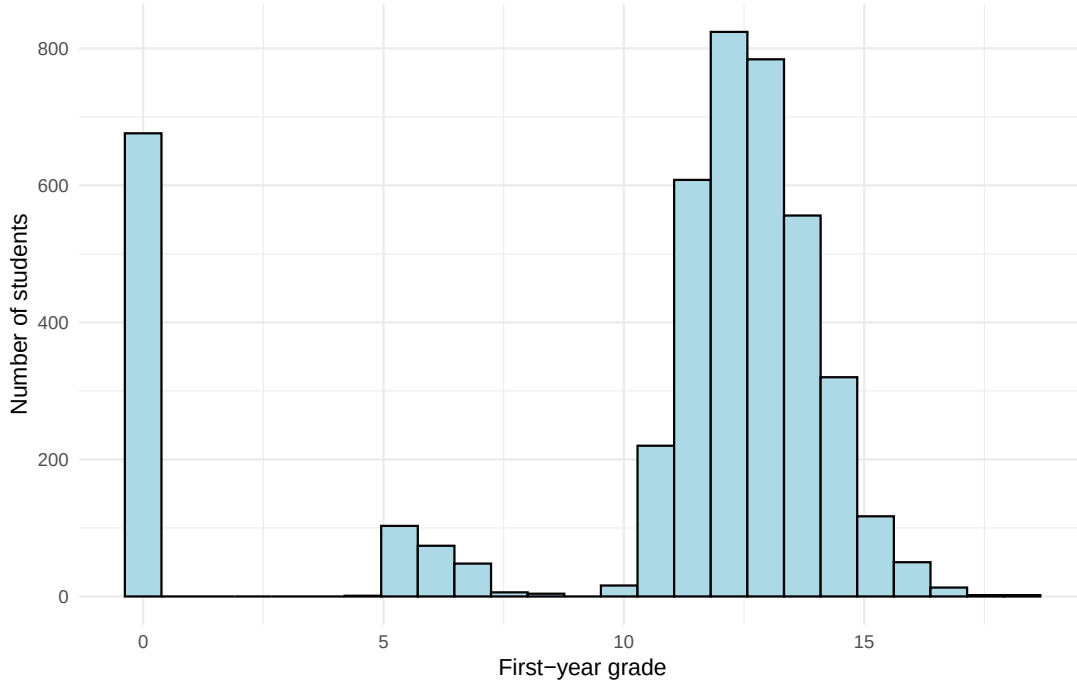
**Figure 1.** Distribution of first-year grade in the full sample.

Table 3. Preliminary model coefficient estimates.

Term	Estimate	Std. Error	t value	p-value
(Intercept)	9.8472	0.7992	12.321	< 0.001
admission_grade	0.0277	0.0059	4.706	< 0.001
previous_qualification_grade	-0.0068	0.0065	-1.042	0.297
age_at_enrollment	-0.0780	0.0094	-8.263	< 0.001
gender1	-1.6550	0.1487	-11.131	< 0.001
scholarship_holder1	1.4604	0.1654	8.830	< 0.001

Section 3: Primary model results and diagnostics. (300 words)

We began with the same preliminary multiple linear regression model as in Part 1:

`grade_year1 ~ admission_grade + previous_qualification_grade + age_at_enrollment + gender + scholarship_holder.`

The preliminary estimates suggested that `admission_grade` was positively associated with first-year academic performance, `age_at_enrollment` was negatively associated with performance, and `scholarship_holder` was positively associated with performance. The coefficient for `previous_qualification_grade` was not statistically significant after adjustment for the other predictors. The indicator for `gender = 1` was negative relative to the reference level `gender = 0`.

However, the residual diagnostics showed that the preliminary model assumptions were not well supported. In the residuals-versus-fitted plot, the residuals displayed a strong downward trend and clear banding rather than a roughly patternless cloud around zero. This gave evidence against linearity and against constant variance. The Q-Q plot showed substantial departures from the reference line, especially in the lower tail, which indicated non-normality. The histogram of standardized residuals was also far from bell-shaped.

The most important feature was a line of extreme negative residuals associated with observations having `grade_year1 = 0`. In other words, the model was trying to explain a response variable with a large structural point mass at zero using a single Gaussian linear-regression equation. That feature created severe misspecification and motivated the model-selection work that followed. Because the sample size is large, these diagnostic patterns are unlikely to be a small-sample artifact, so the need for revision was substantive rather than cosmetic. The coefficient table was still useful as a starting descriptive summary for later valid inference, but it could not justify final inference on its own. Under the course framework, interpretation must follow model adequacy, not come before it. For that reason, Section 3 functions as a bridge: it records what the initial model suggested, then explains why those apparent relationships had to be re-examined using a better-specified modeling strategy in Section 4.

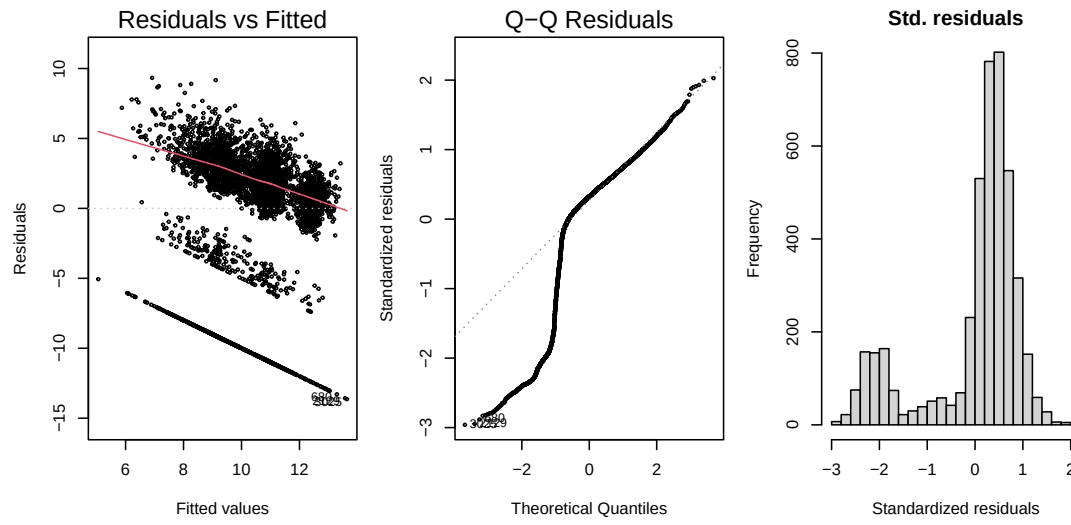


Figure 2. Preliminary model diagnostics.

Section 4: Model selection. (961 words)

The model-selection stage focused on the three questions in the project instructions: whether transformations were needed, whether problematic observations should be removed, and whether any predictors should be excluded from the final model. We followed the course strategy of fitting a broad initial model, checking adequacy, applying justified transformations, comparing candidate specifications on the same scale when appropriate, and then re-checking the refined model.

We began with the preliminary model fit on the full sample because it matched the original research question and included the five predictors supported by the literature. The residuals-versus-fitted plot showed three clear bands of points rather than a random cloud, and the red smooth line sloped downward across fitted values. The Q-Q plot showed major departures from the reference line, especially in the lower left tail. Together, these diagnostics suggested that the linearity and normality assumptions were not well satisfied, and the histogram of standardized residuals reinforced the same conclusion.

To keep students with zero grades in the sample, we first transformed the response. Because the data contain zeros, the usual $\log(y)$ transformation cannot be used on the full sample, so we fit M1 with $\log(\text{grade_year1} + 1)$ as the response. This compressed the residual scale, but the three-band pattern remained and the Q-Q plot showed little improvement. The problem was therefore not only skewness. Instead, zero-grade observations appeared to behave differently from positive-grade observations, so a single Gaussian linear model remained inappropriate even after the transformation.

We then examined whether transforming `age_at_enrollment` helped on the full sample. The descriptive summaries had already shown that age was strongly right-skewed, and the course notes recommend transforming predictors when the linearity assumption is doubtful. Replacing `age_at_enrollment` by $\log(\text{age_at_enrollment})$ produced M2. Relative to M1, the fitted values and residuals were somewhat compressed and the adjusted R-squared increased slightly, but the structural banding tied to zero grades was still present. Since the exact-zero mass was the dominant problem, transforming age alone could not rescue the full-sample model.

We next carried out a sensitivity analysis on the subset of students with `grade_year1 > 0`. This changed the inferential target from all entrants to students with positive first-year grades, but provided a more suitable setting for linear regression. When the raw-response model was fit on this subset, giving M3, the most severe banding disappeared and the residuals looked more randomly scattered around zero. However, some downward drift at low fitted values and lower-tail Q-Q departure remained.

Applying $\log(\text{grade_year1})$ on the positive subset led to M4. This addressed the remaining skewness more effectively. The smooth trend in the residuals-versus-fitted plot became much closer to horizontal, and the Q-Q plot followed the diagonal more closely through the center, although some deviation remained in the lower tail. The information criteria on the same sample and response scale also improved substantially relative to M3, supporting the conclusion from the diagnostics. Because the subset excludes zeros, the ordinary log transformation is now well defined, so M4 was more interpretable and more defensible than the full-sample $\log(y + 1)$ alternative.

We then revisited the age transformation within the positive-grade, log-response framework. Replacing raw age by $\log(\text{age_at_enrollment})$ gave M5. Compared with M4, the diagnostics were similar, but the model with `log_age` had slightly better adjusted R-squared and lower AIC and BIC. This modest but consistent evidence justified retaining `log_age`. Table 4 places the seven

candidate models side by side.

Table 4. Candidate-model comparison used during model selection. These information criteria are most directly comparable for models fit on the same sample and response scale.

Model	n	Adjusted R^2	AIC	BIC
M0: all sample, raw response, raw age	4424	0.0827	26094.13	26138.89
M1: all sample, $\log(y + 1)$, raw age	4424	0.0653	11739.00	11783.76
M2: all sample, $\log(y + 1)$, $\log(\text{age})$	4424	0.0672	11729.84	11774.60
M3: positive subset, raw response, raw age	3748	0.0909	15680.73	15724.33
M4: positive subset, $\log(y)$, raw age	3748	0.0722	-1303.91	-1260.31
M5: positive subset, $\log(y)$, $\log(\text{age})$	3748	0.0745	-1313.09	-1269.49
M6: final chosen model	3748	0.0742	-1313.19	-1275.82

To keep the diagnostic comparison readable, Figure 3 shows four representative models rather than all seven candidates. We omit M1 because M2 is the stronger full-sample transformed version of the same idea. We include M4 because the change from M3 to M4 is the key evidence that the positive-grade subset benefits from the ordinary log transformation. Figure 3 therefore highlights the main steps in the modeling path: the original full-sample model M0, the strongest transformed full-sample alternative M2, the positive-subset log-response model M4, and the final reduced model M6. The more detailed final diagnostics for M6 are then shown separately in Section 5.

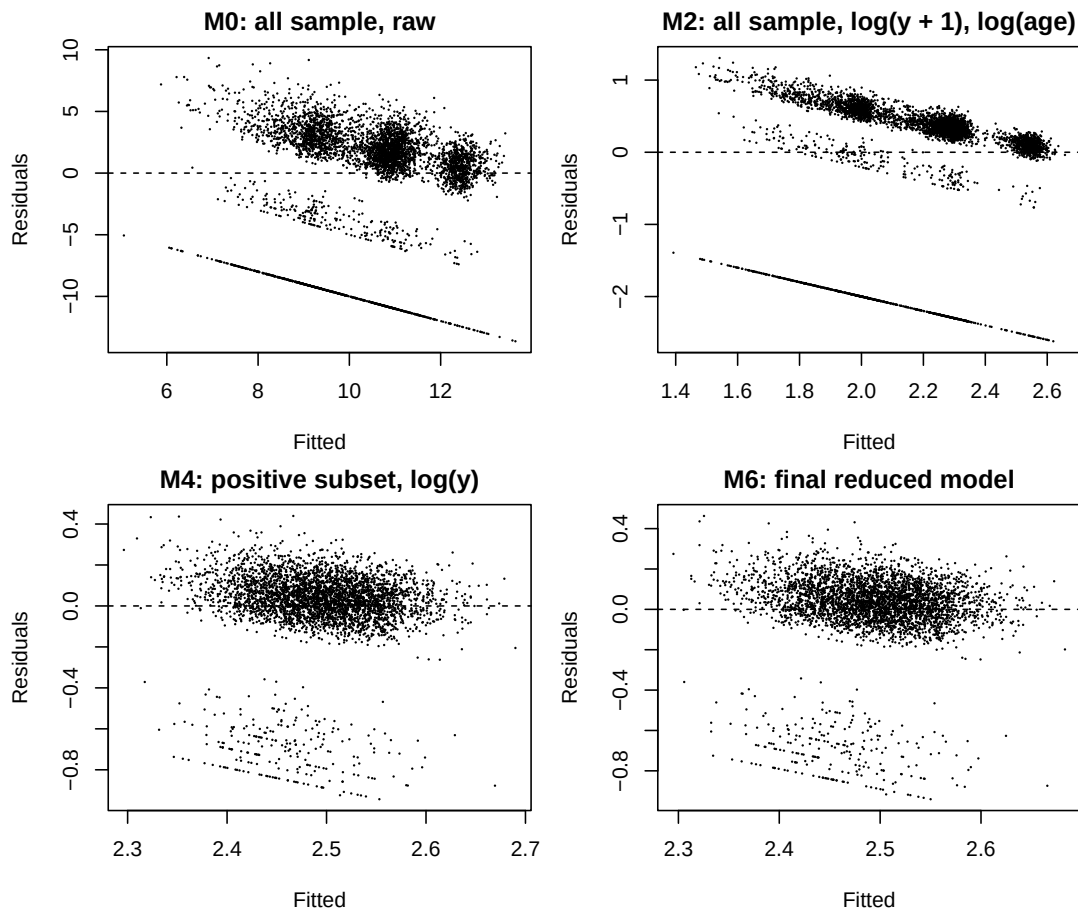


Figure 3. Residual-versus-fitted comparisons for representative candidate models.

With the transformations decided, we turned to variable selection. The key comparison was between M5, which kept all five predictors, and M6, which removed `previous_qualification_grade`. The two models had almost identical explanatory power: adjusted R-squared decreased only from 0.0745 to 0.0742, while AIC became slightly lower and BIC clearly lower for the reduced model. In addition, `previous_qualification_grade` was not statistically significant in M5 and overlaps conceptually with `admission_grade` because both measure prior academic preparation. These results suggest redundancy rather than omission of an essential predictor, so we selected the simpler four-predictor model. To strengthen that decision, we also inspected the best-subset rankings on the positive-grade, log-response scale. The top-ranked models all contained `admission_grade`, `log_age`, `gender`, and `scholarship_holder`, while models that dropped `admission_grade` or `scholarship_holder` performed worse. By contrast, adding `previous_qualification_grade` produced only a negligible gain in adjusted R-squared and did not offset the BIC penalty. The broader subset comparison therefore supported the same conclusion as the direct M5-versus-M6 comparison: M6 preserved the key structure while avoiding an unnecessary predictor.

Table 5. Top subset models on the positive-grade, log-response scale.

Predictors	k	Adjusted R^2	AIC	BIC
admission_grade + log_age + gender + scholarship_holder	4	0.0742	-1313.19	-1275.82
admission_grade + previous_qualification_grade + log_age + gender + scholarship_holder	5	0.0745	-1313.09	-1269.49
admission_grade + previous_qualification_grade + log_age + gender	4	0.0667	-1282.75	-1245.38
admission_grade + log_age + gender	3	0.0664	-1282.38	-1251.24
admission_grade + previous_qualification_grade + gender + scholarship_holder	4	0.0649	-1275.66	-1238.29
admission_grade + gender + scholarship_holder	3	0.0641	-1273.34	-1242.20
admission_grade + previous_qualification_grade + log_age + scholarship_holder	4	0.0627	-1266.61	-1229.23
admission_grade + log_age + scholarship_holder	3	0.0622	-1265.68	-1234.53

Finally, we assessed problematic observations in M6 using studentized residuals, leverage, Cook's distance, DFFITS, and DFBETAS. Many cases were flagged by at least one threshold, which is expected in a large sample because rules such as $4/n$ become very small. The maximum Cook's distance was only 0.0189, far below the more severe benchmark of 1, and the flagged cases were plausible low-performing students rather than obvious recording errors. Influence diagnostics are meant to trigger investigation, not automatic deletion, so no observations were removed. Table 6 summarizes the screening thresholds and the number of cases flagged by each rule. Table 7 then lists the observations with the largest Cook's distances. Figure 4 shows the corresponding influence plots for the final selected model.

Table 6. Influence-summary thresholds and flagged counts for the final model.

Quantity	Value
n	3748.0000
p	4.0000
Leverage cutoff $2(p + 1) / n$	0.0027
Cook's distance cutoff $4 / n$	0.0011
DFFITs cutoff $2 * \sqrt{(p + 1) / n}$	0.0730
DFBETAS cutoff $2 / \sqrt{n}$	0.0327
Count $ \text{studentized residual} > 4$	46.0000
Count leverage $>$ cutoff	239.0000
Count Cook's distance $>$ cutoff	253.0000
Count $ \text{DFFITs} >$ cutoff	253.0000
Count with any $ \text{DFBETAS} >$ cutoff	375.0000
Maximum $ \text{studentized residual} $	4.6510
Maximum leverage	0.0078
Maximum Cook's distance	0.0189
Maximum $ \text{DFFITs} $	0.3084
Maximum $ \text{DFBETAS} $	0.2856

Table 7. Top observations ranked by Cook's distance in the final model.

Grade	Admit	Age	Gen	Schol	Fit	Resid	RStud	Lev	Cook
6.000	183.5	18	0	0	2.666	-0.874	-4.329	0.0051	0.0189
5.000	150.0	48	0	0	2.491	-0.882	-4.364	0.0039	0.0148
5.250	161.8	33	1	0	2.504	-0.846	-4.183	0.0030	0.0104
5.500	153.0	18	1	1	2.576	-0.872	-4.311	0.0028	0.0104
4.900	130.0	36	0	1	2.506	-0.917	-4.536	0.0025	0.0103
6.071	161.1	47	1	0	2.471	-0.668	-3.303	0.0045	0.0098
5.786	156.4	44	1	0	2.464	-0.709	-3.506	0.0037	0.0091
5.000	100.0	25	1	1	2.405	-0.796	-3.936	0.0029	0.0091

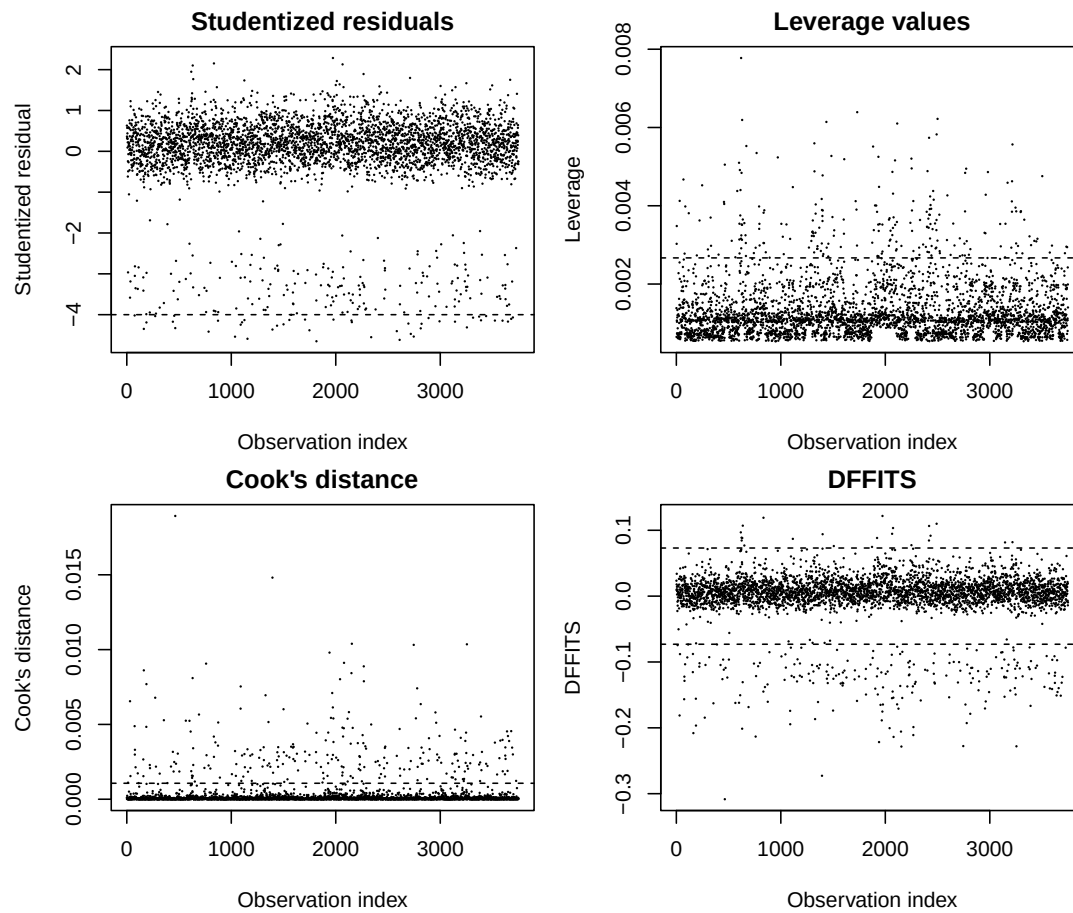


Figure 4. Influence diagnostics for the final model.

Section 5: Final model inference and results. (1012 words)

Having established the final model in Section 4, we now interpret associations between enrollment characteristics and first-year academic performance among students with positive first-year grades. The final model regresses $\log(\text{grade_year1})$ on admission_grade , $\log(\text{age_at_enrollment})$, gender , and $\text{scholarship_holder}$ using the subset with $\text{grade_year1} > 0$. Because a single Gaussian linear model was not adequate for the full sample, the final inferential question is narrower than the original proposal. Table 8 summarizes the final model estimates, including coefficients, standard errors, confidence intervals, t statistics, and p-values. All four retained predictors are statistically significant, and each confidence interval excludes zero. Because the response is log-transformed, all coefficients are interpreted on the log-grade scale.

Table 8. Final model coefficient table with standard errors, confidence intervals, and p-values.

Term	Estimate	Std. Error	CI lower	CI upper	t value	p-value
(Intercept)	2.4215	0.0514	2.3207	2.5222	47.125	< 0.001
admission_grade	0.0027	0.0002	0.0022	0.0032	11.412	< 0.001
$\log_age$	-0.0862	0.0133	-0.1123	-0.0602	-6.483	< 0.001
gender1	-0.0513	0.0073	-0.0656	-0.0371	-7.055	< 0.001
$\text{scholarship_holder1}$	0.0439	0.0076	0.0289	0.0589	5.737	< 0.001

The intercept estimate is 2.4215. Substantively, it is not the most important quantity because it corresponds to unrealistic predictor values, but it is needed to define the fitted regression surface.

The coefficient for admission_grade is positive, about 0.0027, with a 95% confidence interval from 0.0022 to 0.0032. Holding $\log_age$, gender , and $\text{scholarship_holder}$ fixed, a one-point increase in admission_grade is associated with an estimated increase of 0.0027 in expected $\log(\text{grade_year1})$. Among students with positive first-year grades, stronger academic preparation at entry is therefore associated with better first-year performance after adjustment for the other retained characteristics. This matches both the literature and the preliminary model, but here it is supported within a much more defensible specification.

The coefficient for $\log_age$ is negative, about -0.0862, with a 95% confidence interval from -0.1123 to -0.0602. Holding the other predictors fixed, increases in age are associated with lower expected log first-year grade. In context, older entrants in the positive-grade subset tended to perform somewhat less well in the first year after adjustment for prior achievement, gender category, and scholarship status. This should not be read causally; age may proxy differences in educational pathway, time away from school, work responsibilities, or family obligations not measured directly in the dataset.

The indicator coefficient for $\text{gender} = 1$ is about -0.0513, with a 95% confidence interval from -0.0656 to -0.0371. Relative to the reference category $\text{gender} = 0$, students in category 1 have lower expected log first-year grades after controlling for the other predictors. Since the dataset uses numerical coding, we intentionally keep the interpretation at the coded-category level rather than attaching labels that are not necessary for the statistical conclusion. At the same time, the coefficient should be read cautiously. It describes an adjusted association inside this dataset, not a causal explanation of group differences.

The coefficient for $\text{scholarship_holder} = 1$ is about 0.0439, with a 95% confidence interval from 0.0289 to 0.0589. Conditional on admission_grade , $\log_age$, and gender , scholarship recipients

have higher expected log first-year grades than non-recipients. However, the coefficient does not by itself separate financial support from the selection mechanisms used to award scholarships.

The omitted predictor `previous_qualification_grade` also deserves interpretation because its exclusion is part of the final answer. In the five-predictor candidate model, its coefficient was not statistically significant, the adjusted R-squared gain from retaining it was negligible, and moving to the reduced model lowered AIC by about 0.1 and BIC by about 6.33. A partial F-test comparing M6 with M5 was also not significant at the 5% level, with p-value 0.168. Under the course framework, the additional predictor did not reduce residual variation enough to justify keeping it. The most plausible substantive explanation is overlap with `admission_grade`. Both variables capture prior academic achievement, but `admission_grade` appears to be the more informative measure for this response in this dataset.

The final model results are broadly consistent with the literature reviewed in Section 1. In particular, the positive `admission_grade` coefficient is consistent with prior work emphasizing the importance of academic preparation at entry (Wolniak and Engberg, 2010), while the negative `log_age` estimate and adjusted gender difference are in line with studies showing that age and gender can be associated with first-year academic outcomes (Newman-Ford et al., 2009). The positive `scholarship_holder` coefficient also points in the same direction as earlier evidence that financial support is associated with stronger student performance and persistence (Stater, 2009). The main difference from the literature-based starting specification is the exclusion of `previous_qualification_grade`, which in this dataset adds little explanatory value once `admission_grade` is already in the model.

The final fitted equation was:

$$\widehat{\log(\text{grade_year1})} = 2.4215 + 0.00269 \text{ admission_grade} - 0.08624 \log(\text{age_at_enrollment}) \\ - 0.05135 I(\text{gender} = 1) + 0.04387 I(\text{scholarship_holder} = 1).$$

Model performance metrics help place these coefficient results in perspective. Table 9 reports the summary fit statistics for the final model. The final model has R-squared about 0.0752 and adjusted R-squared about 0.0742, so only a modest proportion of the variation in `log(grade_year1)` is explained by the retained enrollment characteristics. In educational data, many important determinants of performance are unmeasured at enrollment, so the model is better viewed as an interpretable descriptive model than as a high-accuracy predictive tool.

Table 9. Final model performance metrics.

Metric	Value
R-squared	0.0752
Adjusted R-squared	0.0742
AIC	-1313.1921
BIC	-1275.8182
Residual standard error	0.2029
RMSE	0.2028
Sample size	3748.0000

The information criteria tell a similar story about adequacy and parsimony. These values are

directly comparable to the five-predictor positive-subset log model, and within that comparison the reduced model is preferred because it achieves nearly the same adjusted R-squared while improving both AIC and BIC. The residual standard error and RMSE are both about 0.203 on the log scale. Taken together, the coefficient table, confidence intervals, literature comparison, and model-performance metrics provide a coherent answer to the refined research question.

The final diagnostics still show some lower-tail departure and residual structure, but inference from this restricted model is more credible than from the original full-sample specification. Figure 5 is placed here because it provides the graphical evidence for that improved-but-imperfect adequacy statement.

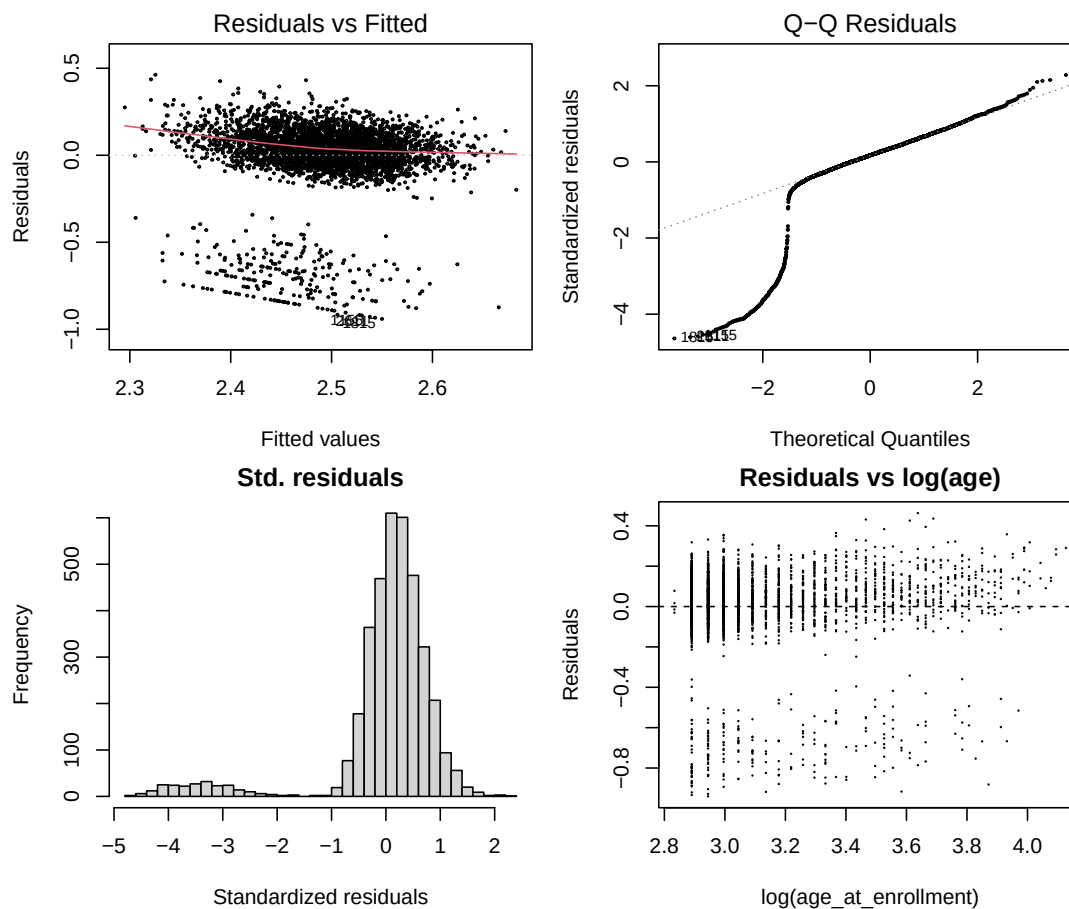


Figure 5. Final model diagnostics.

Section 6: Discussion and conclusion. (1115 words)

This project examined how first-year academic performance is associated with characteristics observed at enrollment, but the main conclusion is both methodological and substantive. Methodologically, the analysis shows that the original full-sample Gaussian linear model was not appropriate for this response. The large exact-zero mass in `grade_year1` created severe misspecification, visible in the residual banding, the strong lower-tail departure in the Q-Q plot, and the non-bell-shaped standardized residual histogram from the preliminary model. A linear model is useful only when its fitted mean structure and residual behavior are plausible enough to support interpretation.

For that reason, the final model was fit on the subset of students with `grade_year1 > 0` using `log(grade_year1)` as the response and `log(age_at_enrollment)` as a transformed predictor. This decision narrowed the inferential target, but it also produced a more credible framework for analysis. In practical terms, the report therefore gives a qualified answer to the research question: among students with positive first-year grades, enrollment-time characteristics do show meaningful adjusted associations with academic performance.

A first key finding is the positive association between `admission_grade` and first-year performance. The final coefficient is positive, statistically significant, and estimated with a narrow confidence interval that excludes zero. Even after controlling for age, gender category, and scholarship status, students entering with stronger admission records tended to earn higher first-year grades. This result is consistent with the literature reviewed earlier. At the same time, the effect size is modest on the log scale, so admission performance is informative rather than close to a complete explanation of first-year success.

A second key finding is the negative association between `log_age` and `log(grade_year1)`. The coefficient is below zero, its confidence interval remains entirely negative, and the sign is stable across the more defensible positive-subset models. This suggests that older students in this dataset tended to have somewhat lower first-year grades after adjustment for the other retained predictors. The evidence should not be read causally. More likely, age is acting as a proxy for differences in educational trajectory, work commitments, caregiving responsibilities, or time away from formal study. Even so, the negative adjusted association is practically meaningful because it identifies a group that may face a different transition into university.

A third key finding is the positive association between `scholarship_holder` status and performance. The final coefficient is positive, statistically significant, and supported by a confidence interval entirely above zero. Conditional on the other predictors, scholarship recipients tended to have higher expected log first-year grades than non-recipients. Scholarship receipt may reflect both financial support and selection into a group of students who were already especially prepared or motivated.

A fourth key finding is that the gender indicator remained in the final model with a negative coefficient for category 1 relative to category 0. The estimate is statistically precise enough that sampling variation alone is not a convincing explanation for the observed adjusted difference. However, interpretation should remain limited to the coded categories in the dataset. Without additional contextual information, the coefficient does not tell us why the groups differ or whether the difference would persist under richer adjustment. What it does show is that demographic differences remained associated with academic performance even after controlling for the other retained enrollment variables.

The final model also provides an important negative result: `previous_qualification_grade` was not

retained. This was not because the variable was theoretically meaningless. It was dropped because the evidence indicated that it added very little once `admission_grade` was already in the model. The adjusted R-squared gain was negligible, AIC and BIC favored the simpler model, and the partial F-test comparing the larger and smaller positive-subset log models was not significant. In substantive terms, the two achievement measures appear to overlap, and `admission_grade` seems to carry the more useful information for this response. This improves interpretability because the final model keeps the clearer signal without unnecessary redundancy.

Several recommendations follow from these findings. Enrollment-time variables such as `admission_grade`, `age_at_enrollment`, and `scholarship_holder` may be useful for identifying academic risk patterns early in the student pathway. In particular, students with weaker admission records and older entrants may warrant additional monitoring or proactive advising during the first year. Scholarship-related support also appears connected to stronger outcomes, so institutions may want to examine whether assistance or advising attached to scholarship programs could be extended more broadly. Because the explained variation is low, the model is better suited to broad planning and screening than to deterministic prediction about any one student.

These recommendations are relevant precisely because the model's limitations are taken seriously. The final adjusted R-squared is only about 0.074, which means that most student-to-student variation remains unexplained. First-year performance depends on many factors that are not captured here, including study habits, attendance, teaching quality, program difficulty, peer networks, mental health, family support, and financial stress outside the scholarship indicator. The model therefore should not be treated as a strong predictive engine. Its value lies in identifying adjusted patterns that are statistically supported and institutionally interpretable.

The analysis also has several important limitations beyond low explanatory power. First, the data come from a single Portuguese institution, so external validity is limited. Second, the final inferential model applies only to students with positive first-year grades. Students with zero grades are substantively important, but the evidence in this report suggests that they are not well represented by the same Gaussian linear mechanism. Third, some variables are available only as coded categories, which constrains interpretation. Fourth, the project intentionally used methods close to the course material, so the analysis does not attempt more specialized models that might fit the response structure better.

Future work could improve the analysis in several directions. One extension would be to model the zero-grade group separately from the positive-grade group, since the diagnostics suggest that these groups arise from different processes. A second improvement would be to add more predictors from the broader dataset, especially variables related to program, socioeconomic background, or student circumstances at entry. A third would be to examine course-appropriate interactions, such as whether the admission-grade association differs by scholarship status or gender category. A fourth would be to explore program-level heterogeneity, since students in different academic programs may face different grading structures and transition challenges.

In conclusion, this report provides a cautious but meaningful answer to the refined research question. Among students with positive first-year grades, higher `admission_grade` and `scholarship_holder` status are associated with higher academic performance, while higher `age_at_enrollment` and membership in gender category 1 are associated with lower performance after adjustment. More broadly, the report demonstrates that careful model specification is essential. The strongest conclusion is not only about which predictors matter, but also about why diagnostic evidence must guide the final inferential model in applied regression work.

Section 7: Author contributions.

Emilia Li was responsible for the coding and analysis for Section 4: Model Selection. She carried out the model-selection work, including transformation checks, diagnostic assessment, comparison of candidate models, and supporting code for the final report.

Shimin Zeng and Felix Zhao were responsible for Section 5: Final Model Inference and Results. They developed the final-model results section, including coefficient interpretation, statistical inference, and presentation of the final model findings.

Jingxiu Zeng was responsible for Section 6: Discussion and Conclusion. She wrote the discussion of the main findings, limitations, implications, and concluding remarks based on the final model.

Siqi Yang and Shimin Zeng were responsible for the Final Report as a whole, including Sections 1, 2, and 3. They wrote Section 1: Introduction, Section 2: Data Description, and Section 3: Primary Model Results and Diagnostics.

Siqi Yang was also responsible for the final integration of the project, including compiling all sections, standardizing formatting, titles, figure and table numbering, references, checking word counts and rubric requirements, and preparing the final submission version.

Section 8: References.

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